Reg. No. | | | | | | | | | | | [14EE62]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE-2019

DIGITAL SIGNAL PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions, choosing at least two from each part.
2. Normalized filter tables are allowed.

PART-A

1. a. Prove the circular convolution property of Discrete Fourier Transform (DFT). **04**

   b. Compute the circular convolution using DFT and IDFT for the following sequences: $x_1(n)=\{2,3,1,1\}$ and $x_2(n)=\{1,3,5,3\}$. **06**

   c. X(k) is a finite length sequence given as $X(k) = \{0, 1 + j, 1, 1 - j\}$. Using the properties of DFT, find DFTs of the following sequences: **04**

      i. $x_1(n) = e^{(jn\,n)/2}\, x(n)$,

      ii. $x_2(n) = (x(n-1))_4$

2. a. Obtain the Linear Convolved output y(n)=x(n)*h(n) using circular convolution, given that: $x(n) = \{1, 1, 0, -1, -1\}$ and $h(n)=\{1, 2, 3, 2, 1\}$. **06**

   b. Using **OVERLAP ADD** method, determine the output y(n) of a system whose impulse response is **h(n)={1, 1, 1}** and input is **x(n)={3, -1, 0, 1, 3, 2, 0, 1, 2, 1}.** Use 5 point circular convolution. **08**

3. a. Compute the DFT of the sequence $\mathbf{x(n)=(-1)^n}$ for the period N=8 using DIT-FFT algorithm. **10**

   b. Show that FFT method is more computationally efficient than Direct computation of DFT. **04**

4. a. X(k) is a 8-point DFT of a sequence x(n), the first five samples of X(k) are given as, X(0)=0, X(1)=2+j2, X(2) = -j4, X(3) = 2-j2, X(4)=0. Determine the remaining samples of X(k). Find the original sequence x(n) using Decimation – in – frequency (DIF) FFT algorithm. **08**

   b. Develop DIT-FFT algorithm for N=9 = 3 x 3 and draw the complete signal flow graph. **06**

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PART-B

5. **a.** A system is represented by a transfer function H(z) as given by: **08**

$$H(z) = \frac{1 - z^{-1}}{1 - 0.2z^{-1} - 0.15z^{-2}}$$

i. Find the difference equation.

ii. Draw the Direct form I, Direct form II and cascade realization of the IIR system.

b. Obtain the direct form and cascade form realization of the FIR system function: **06**

$H(z) = (1 + 3z^{-1} - z^{-3} - z^{-4})$

6. **a.** The system function of the analog filter is given as: $H_a(s) = \frac{s+0.1}{(s+0.1)^2+9}$. Obtain the system function of the IIR digital filter using Impulse Invariance method for $\omega_r = \frac{\pi}{2}$. **04**

b. Design a lowest order Digital Butterworth filter satisfying following constraints: **10**

$\sqrt{0.5} \leq |H(e^{j\omega})| \leq 1$ for $0 \leq \omega \leq \frac{\pi}{2}$

$|H(e^{j\omega})| \leq 0.2$ for $\frac{3\pi}{4} \leq \omega \leq \pi$

With T=1s, apply impulse invariant transformation.

7. **a.** The system function of the analog filter is given as $H_a(s) = \frac{s+0.1}{(s+0.1)^2+16}$. Obtain the system function `n of the digital filter using bilinear transformation which is resonant at $\omega_r = \frac{\pi}{2}$. **04**

b. Design a digital filter using Chebyshev approximation and bilinear transformation to meet the following specifications: **10**

Passband Ripple of 1dB for $0 \leq \omega \leq 0.15\pi$

Stopband attenuation $\geq 20dB$ for $0.45\pi \leq \omega \leq \pi$

8. **a.** Design a normalized linear phase FIR filter having the phase delay of $\tau = 4$ and at least 40 dB attenuation in the stop band. Also obtain the magnitude and frequency response of the filter. **08**

b. Design a low pass FIR filter using frequency sampling technique having cutoff frequency of $\frac{\pi}{2}$ rad/sample. The filter should have linear phase and length of 17. **06**